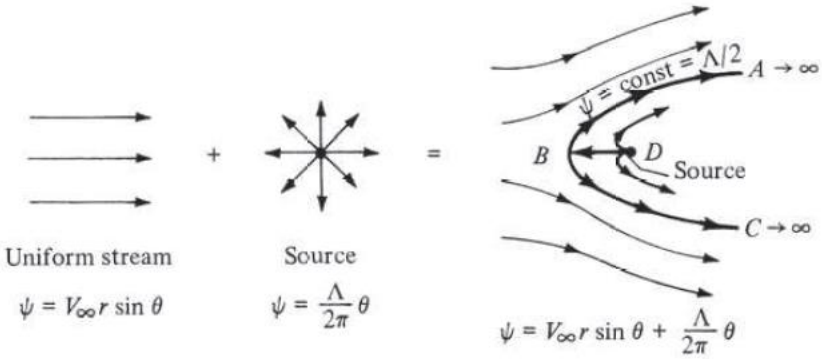
Combination of Elementary Flows

Summary of stream function and velocity potential:

Type of flow	Velocity	ϕ	ψ
Uniform flow in x direction	$u = V_\infty$	$V_\infty x$	$V_\infty y$
Source	$V_r = \dfrac{\Lambda}{2\pi r}$	$\dfrac{\Lambda}{2\pi}\ln r$	$\dfrac{\Lambda}{2\pi}\theta$
Vortex	$V_\theta = -\dfrac{\Gamma}{2\pi r}$	$-\dfrac{\Gamma}{2\pi}\theta$	$\dfrac{\Gamma}{2\pi}\ln r$
Doublet	$V_r = -\dfrac{\kappa}{2\pi}\dfrac{\cos\theta}{r^2}$	$\dfrac{\kappa}{2\pi}\dfrac{\cos\theta}{r}$	$-\dfrac{\kappa}{2\pi}\dfrac{\sin\theta}{r}$
	$V_\theta = -\dfrac{\kappa}{2\pi}\dfrac{\sin\theta}{r^2}$		

Flow over the semi-infinite body:

- Superposition of uniform flow and a source



Superposition of uniform flow and a source

Stream function:

$$\boldsymbol{\Psi} = \text{Uniform} + \text{Source}$$

$$\psi = \psi_1 + \psi_2 = U_\infty r \sin\theta + \frac{\Lambda}{2\pi}\theta$$

Once we have the stream function, we can take its spatial derivatives to get the velocity field.

Keep in mind that we are in cylindrical coordinates.

$$u_r = \frac{1}{r}\frac{\partial\psi}{\partial\theta} = U_\infty \cos\theta + \frac{\Lambda}{2\pi r}$$

$$u_\theta = -\frac{\partial\psi}{\partial r} = -U_\infty \sin\theta$$

In these flows, the stagnation points are important because they often lie on the streamline that represents a body's boundary for that particular flow.

The radial velocity from a source is $\frac{\Lambda}{2\pi r}$ and the component of freestream velocity in the radial direction is $U_\infty \cos\theta$.

At the stagnation point,

$$u_r = U_\infty \cos\theta + \frac{\Lambda}{2\pi r} = 0$$
$$u_\theta = U_\infty \sin\theta = 0$$

Solving for *r* and *θ*, we find that one stagnation point exists, located at (r, θ)

$$(r,\theta) = \left(\frac{\Lambda}{2\pi U_\infty}, \pi\right)$$

The coordinate for the stagnation point and the flow is at a θ angle of π or 180 degrees and a distance from the origin of $(\Lambda/2\pi U_\infty)$. **Note: Origin is defined by the center of the source.**

Boundary: The streamline that has a stagnation point is found by taking the stream function and evaluating it at the coordinates (r, θ).

Stream function at $(r,\theta) = \left(\frac{\Lambda}{2\pi U_\infty}, \pi\right)$ can be written as,

$$\psi = U_\infty \left(\frac{\Lambda}{2\pi U_\infty}\right)\sin(\pi) + \frac{\Lambda}{2\pi}\pi = \frac{\Lambda}{2\pi} = \text{constant}$$

Where $\sin(\pi) = 0$

This stream function represents the streamline with the boundary on it.

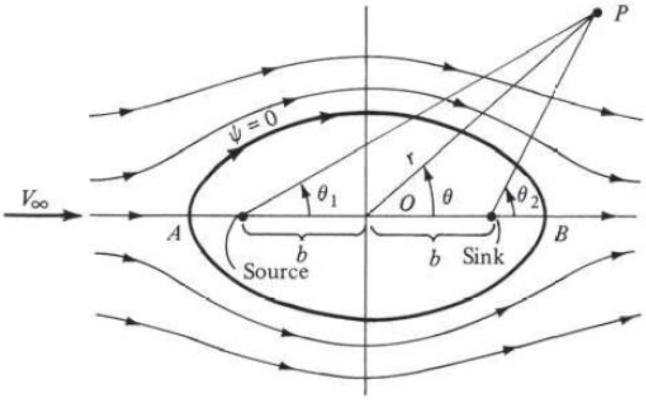
Flow over Rankine oval:

- Superposition of uniform flow, a source, and a sink.
- This problem was first solved in the nineteenth century by the famous Scottish engineer W. J. M. Rankine; hence, the shape given is called Rankine oval.

Stream function:

$$\psi = \text{uniform} + \text{Source} + \text{Sink} = \psi_1 + \psi_2 + \psi_3$$

$$\psi = U_\infty r\sin\theta + \frac{\Lambda}{2\pi}\theta_1 - \frac{\Lambda}{2\pi}\theta_2 = U_\infty r\sin\theta + \frac{\Lambda}{2\pi}(\theta_1 - \theta_2)$$



Superposition of a uniform flow and a source-sink pair

- Now the fact that the source and sink are off from the origin will present some coordinate difficulties.
- Source and sink have different angles and radial locations relative to the point in space.
- Since source and sink are not at the origin, θ_1 and θ_2 are the functions of the azimuthal, radial coordinate from the true origin (r, θ), and the distance of source/sink from the origin.

$$\theta_1 \text{ and } \theta_2 = f(\theta, r, b)$$

$$\theta_1 = \tan^{-1}\left(\frac{r\sin\theta}{r\cos\theta + b}\right) \text{ and } \theta_2 = \tan^{-1}\left(\frac{r\sin\theta}{r\cos\theta - b}\right)$$

To define the velocity field, we need to incorporate these angle functions into the stream function in order to take the accurate derivatives. This will make the problem so complex and difficult to solve. We rarely need the velocity field so we can skip that.

Stagnation:

If a streamline represents the flow around an oval we can expect that the stagnation will happen at the upstream and downstream tips of the body.

Stagnation happens at

$\theta = \theta_1 = \theta_2 = \pi$ **at the leading edge**

$\theta = \theta_1 = \theta_2 = 0$ **at the trailing edge**

$$\psi = U_\infty r\sin\theta + \frac{\Lambda}{2\pi}(\theta_1 - \theta_2) = 0$$

$$\psi = 0$$

This constant represents the streamline that represents the body because it has stagnation points on it.

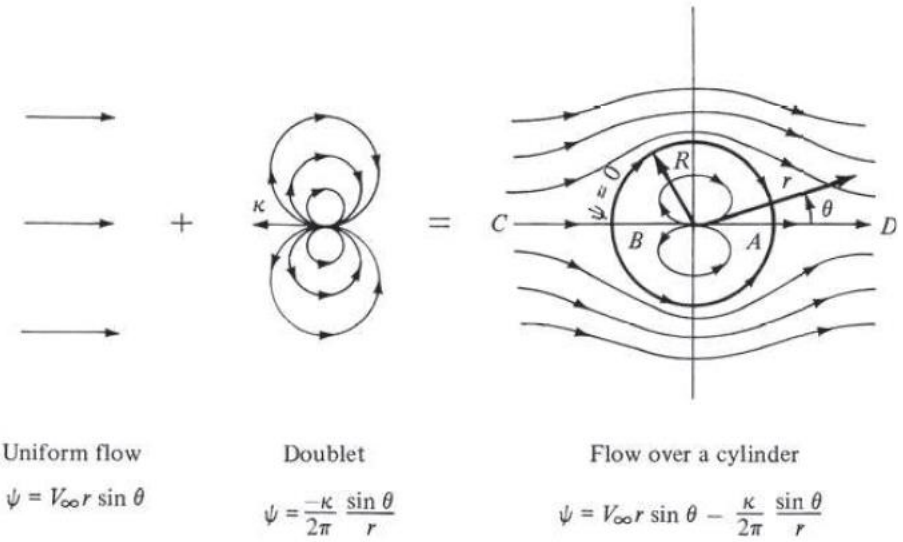
Flow over stationary (non-lifting cylinder):

- Superposition of uniform flow and a doublet.

Stream function:

$$\psi = U_\infty r\sin\theta - \frac{k}{2\pi}\frac{\sin\theta}{r} = U_\infty r\sin\theta\left(1 - \frac{k}{2\pi U_\infty r^2}\right) = U_\infty r\sin\theta\left(1 - \frac{R^2}{r^2}\right)$$

Let $R^2 \equiv k/2\pi U_\infty$



Superposition of a uniform flow and a doublet

Velocity field:

- Taking the spatial derivative of the stream function.

$$u_r = \frac{1}{r}\frac{\partial\psi}{\partial\theta} = \frac{1}{r}(U_\infty r\cos\theta)\left(1 - \frac{R^2}{r^2}\right) = \left(1 - \frac{R^2}{r^2}\right)U_\infty\cos\theta$$

$$u_\theta = -\frac{\partial\psi}{\partial r} = -\left[(U_\infty r\sin\theta)\frac{2R^2}{r^3} + \left(1 - \frac{R^2}{r^2}\right)(U_\infty\sin\theta)\right] = -\left(1 + \frac{R^2}{r^2}\right)U_\infty \operatorname{Sin}\theta$$

Stagnation Points:

To locate the stagnation points, set the radial and tangential velocity to zero.

$$u_r = \left(1 - \frac{R^2}{r^2}\right) U_\infty \cos\theta = 0$$

$$u_\theta = \left(1 + \frac{R^2}{r^2}\right) U_\infty \operatorname{Sin}\theta = 0$$

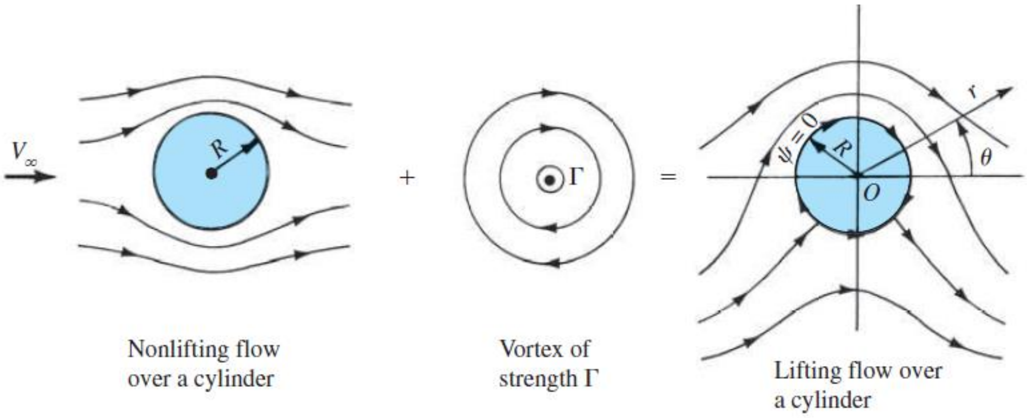
Simultaneously solving Equations for *r* and *θ*, we find that there are two stagnation points, located at *(r, θ)* = *(R,* 0*)* and *(R, π)*.

Plugging the values of r and θ in the stream function, we get

$$\psi = 0$$

$\psi = U_\infty r \sin\theta \left(1 - \frac{R^2}{r^2}\right) = 0$ represent the streamline of the boundary.

Flow over rotating cylinder:



Superposition of non-lifting flow over cylinder and vortex

The stream function for non-lifting flow over a circular cylinder of radius *R* is given by

$$\psi_1 = U_\infty r \sin\theta \left(1 - \frac{R^2}{r^2}\right)$$

The stream function for a vortex of strength Γ is given by

$$\psi_2 = \frac{\Gamma}{2\pi} \ln r + \text{constant}$$

Since the value of the constant is arbitrary, let

$$\text{constant} = -\frac{\Gamma}{2\pi} \ln R$$

So, the stream function for the vortex becomes

$$\psi_2 = \frac{\Gamma}{2\pi}\ln\frac{r}{R}$$

The resulting stream function for the flow

$$\psi = \psi_1 + \psi_2 = U_\infty r\sin\theta\left(1-\frac{R^2}{r^2}\right) + \frac{\Gamma}{2\pi}\ln\frac{r}{R}$$

Velocity field:

$$u_r = \frac{1}{r}\frac{\partial\psi}{\partial\theta} = \frac{1}{r}(U_\infty r\cos\theta)\left(1-\frac{R^2}{r^2}\right) = \left(1-\frac{R^2}{r^2}\right)U_\infty\cos\theta$$

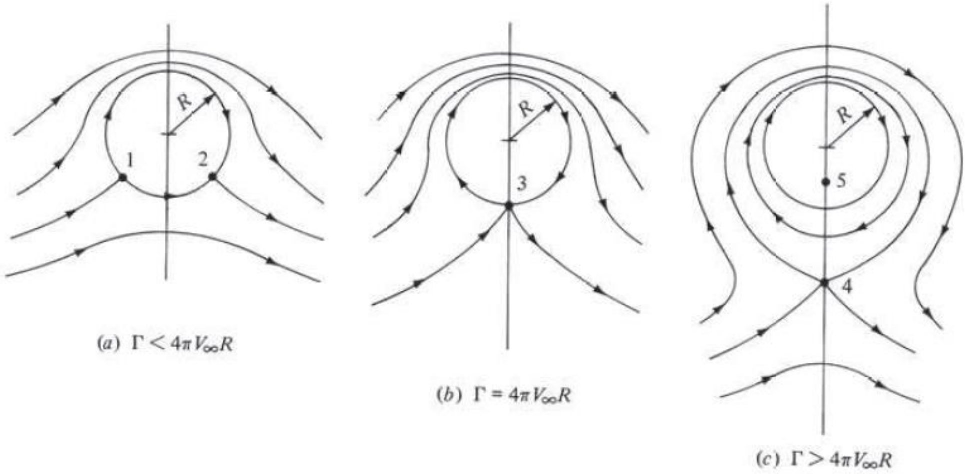
$$u_\theta = -\frac{\partial\psi}{\partial r} = -\left(1+\frac{R^2}{r^2}\right)U_\infty \operatorname{Sin}\theta - \frac{\Gamma}{2\pi r}$$

Stagnation point:

To locate the stagnation points in the flow, set the radial and tangential velocity to zero.

$$u_r = \left(1-\frac{R^2}{r^2}\right)U_\infty\cos\theta = 0$$

$$u_\theta = -\left(1+\frac{R^2}{r^2}\right)U_\infty \operatorname{Sin}\theta - \frac{\Gamma}{2\pi r} = 0$$



Stagnation points for the lifting flow over the cylinder

Substitute r = R and solve for θ, we obtain

$$\theta = \sin^{-1}\left(-\frac{\Gamma}{4\pi U_\infty R}\right)$$

Since Γ is a positive number, θ must be in the third and fourth quadrants. That is, there can be two stagnation points on the bottom half of the circular cylinder, as shown by points 1 and 2.

However, this result is valid only when $\Gamma/4\pi U_\infty R < 1$.
If $\Gamma/4\pi U_\infty R > 1$, there is only one stagnation point on the surface of the cylinder, namely, point $(R, -\pi/2)$.

Substituting $\theta = -\pi/2$ and solving for r we have

$$r = \frac{\Gamma}{4\pi U_\infty} \pm \sqrt{\left(\frac{\Gamma}{4\pi U_\infty}\right)^2 - R^2}$$

In practise,

- Elementary flows are most commonly used in CFD.
- Many flow solving techniques have elementary flows built into them.

Reference:

Anderson, J. (2011)*: Fundamentals of Aerodynamics (SI units)*. McGraw hill.